

Markscheme

Mathematics: analysis and approaches

Practice question bank

29. (a)

Consider the function $f(x) = \frac{3x^2 + 8x - 59}{(x-1)(x+2)(x-5)}$.

Write $f(x)$ as the sum of partial fractions in the form $\frac{A}{x-1} + \frac{B}{x+2} + \frac{C}{x-5}$.

attempt to write $A(x+2)(x-5) + B(x-1)(x-5) + C(x-1)(x+2)$ (M1)

$$3x^2 + 8x - 59 = A(x+2)(x-5) + B(x-1)(x-5) + C(x-1)(x+2) \quad \text{A1}$$

$$\text{substituting } x = 1: -48 = -12A \Rightarrow A = 4 \quad \text{A1}$$

$$\text{substituting } x = -2: -63 = 21B \Rightarrow B = -3 \quad \text{A1}$$

$$\text{substituting } x = 5: 56 = 28C \Rightarrow C = 2 \quad \text{A1}$$

[5 marks]

(b)

Hence find the exact value of $\int_6^7 f(x) dx$.

$$\int f(x) dx = 4 \ln |x-1| - 3 \ln |x+2| + 2 \ln |x-5| \quad \text{(M1)}$$

$$\text{evaluating between } x = 6 \text{ and } x = 7 \text{ gives } \approx 1.762 \quad \text{A1}$$

[2 marks]